

4-12-18 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov – Dec 2018

Max. Marks: 100

Class: SY B.Tech

Name of the Course: Data Communication and Networking

Course Code: UITC402

Duration: 03 Hrs

Semester: IV

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Define Topology. Draw neat diagram for Bus and Ring Topologies and compare them (five points). What are the other topologies?	10
Q 1 (b)	Write the use of the following devices and mention at which layer of ISO/OSI reference model they are used. a) Switch b) Router c) Hub d) Bridge e) Repeater	10
Q2 (a)	List five line coding schemes and explain the Polar line coding schemes in detail with example. OR Define analog to analog conversion? Explain all three conversion methods with suitable example.	10
Q2 (b)	Explain any two guided medium and three unguided medium with their advantages and disadvantages.	10
Q3 (a)	List error detection and correction methods and Explain CRC in detail with suitable example. OR Explain Go-Back-N ARQ and Selective Repeat ARQ in detail.	10
Q3 (b)	A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces (a) 1000 frames per second (b) 500 frames per second (c) 250 frames per second OR	10

	Explain the following:- a) HDLC and b) Hamming code	
Q4 (a)	What are the three main functions performed by network layer? What is routing? Explain Distance vector routing with an example.	10
Q4 (b)	Draw IPV4 Header diagram and differentiate between IPV4 V/S IPV6. OR What is traffic shaping? Explain two methods of traffic shaping in detail.	10
Q5 (a)	Compare the TCP and UDP protocols (any four points). Define the six control bits or flags in TCP. What is their purpose?	10
Q5 (b)	What is congestion control? How it is different from flow control? Explain various congestion prevention techniques.	10

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End Semester Exam
Nov – Dec 2018

Max. Marks: 100

Class: SY BTech

Name of the Course: Analysis of Algorithms

Course Code: UITC403

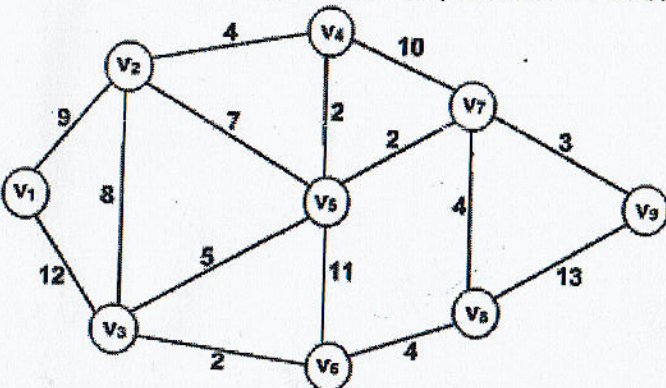
Duration: 3hrs

Semester: IV

Branch: Information Technology

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary justify the same
- (4) Use PEN for writing final answers

Question no.		Max. Marks
Q 1(a)	What is the need of analysis of an algorithm? Explain RAM model in detail. OR Explain asymptotic notations of algorithm analysis.	10
Q 1(b)	Solve following recurrence using substitution and recursion tree methods $T(n) = T(n/3) + T(2n/3) + cn$	10
Q 2 (a)	Explain randomized version of quick sort and derive its time complexity. OR Write pseudo code for merge sort and derive its big-o, big-omega and big-theta time complexity.	10
Q2 (b)	Sort following numbers in descending order using shell sort and radix sort(show all passes). 50, 31, 71, 38, 77, 81, 12, 33, 103, 27	10
Q3 (a)	List the algorithms to find minimum cost spanning tree. Write the algorithm for any two of them and apply any one of two on following to find minimum cost spanning tree.(consider V1 as starting vertex). 	10
Q3 (b)	Explain job sequencing with deadline for the given instances - $n=5$, $\{p1,p2,p3,p4,p5\}=\{20,15,10,5,3\}$ and $\{d1,d2,d3,d4,d5\}=\{2,2,1,3,3\}$	10
Q4 (a)	Find an optimum solution to the knapsack instance $n=7$ and $m=15$ Profit= $\{10,5,15,7,6,18,3\}$ and weight= $\{2,3,5,7,1,4,1\}$	10
Q4 (b)	Write and explain sum of subsets for $n=5$, $w=\{2,7,8,9,15\}$ $m=17$	10

Q5	Short note on (any two) i) 8-queens ii) Travelling sales man problem and solutions iii) Longest common subsequence algorithm iv) Optimal binary search tree	20
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End Semester Exam

Nov - Dec 2018

Max. Marks: 100

Class: SY B Tech

Name of the Course: Computer Organization and Architecture

Course Code: UITC404

Duration: 3 hours

Semester: IV

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain bus interconnection structure with the help of data, address and control Lines	10
Q 1 (b)	Explain in brief Arithmetic Logic Unit and its Hardware implementation.	10
Q2 (a)	Explain Booth's Algorithm for 2's complement Multiplication using Flow chart. And Multiply 7 and 5 using Booths algorithm. (Register size= 4 bits)	10
Q2 (b)	Represent $(178.375)_{10}$ in single and double precision floating point format. (Show stepwise solution) OR Perform division of following numbers using restoring division algorithm. Dividend: 1100, Divisor : 0100 (Show step wise solution)	10
Q3 (a)	Write a note on architecture of 8085 microprocessor. OR Explain any five addressing modes. Give an example of each mode.	10
Q3 (b)	Draw and explain Instruction Cycle state diagram. OR Write a note on micro-programmed control unit.	10
Q4 (a)	Design a two way set associative memory with the following details. Cache consists of 32 words. Each block consists of 4 words Size of main memory: 256 words Also find the number of bits in each of TAG, BLOCK/SET and WORD OR Explain page replacement algorithm. Find out page fault for following string using LRU method. Consider page frame size = 3 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1	10
Q4 (b)	Explain with flowchart i) Interrupt driven I/O ii) Programmed I/O.	10
Q5 (a)	Write a note on pipelining. Show that K stage pipelined processor has K times speedup compared to non-pipelined processor.	10
Q5 (b)	Write a note on Flynn's Classification.	10

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End Semester Examinations

Nov – Dec 2018

Max. Marks: 100

Class: S.Y. B.Tech.

Name of the Course: Web Programming -I

Course Code: UITC405

Duration: 3 Hours

Semester: IV

Branch: I.T.

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks														
Q 1 (a)	Explain the following in brief: 1. n-tier architecture of web application 2. Domain Name System	10														
Q 1 (b)	What are the common design issues faced during web development?	10														
Q2 (a)	1. Write HTML code to draw following table: <i>A test table with merged cells</i><table><tr><td rowspan="2"></td><td colspan="2">Average</td><td rowspan="2">Red eyes</td></tr><tr><td>height</td><td>weight</td></tr><tr><td>Males</td><td>1.9</td><td>0.003</td><td>40%</td></tr><tr><td>Females</td><td>1.7</td><td>0.002</td><td>43%</td></tr></table> 2. Explain different types of lists in HTML with suitable example.		Average		Red eyes	height	weight	Males	1.9	0.003	40%	Females	1.7	0.002	43%	05
	Average		Red eyes													
	height	weight														
Males	1.9	0.003	40%													
Females	1.7	0.002	43%													
Q2 (b)	Explain input elements newly introduced in HTML5 with example. OR Using HTML5, draw a canvas containing Star as shown below. 	10 10														
Q3 (a)	Explain CSS3 animation properties with suitable example. OR Explain CSS Box model with suitable example for margins, paddings and borders.	10 10														
Q3 (b)	Explain CSS3 transition properties with suitable example.	10														
Q4 (a)	Write a JavaScript program to change background color continuously. OR Write a JavaScript program to display weekdays using Switch Case.	10 10														
Q4 (b)	Explain properties, methods and comparison operators for the String Object in JavaScript. OR What is jQuery? Illustrate the use of jQuery for Form Validation.	10 10														
Q5 (a)	Compare XML DTD with XML Schema.	10														
Q5 (b)	Create a XML file to display information about 5 books. Also apply CSS script to this XML file.	10														